EthnoHack 2026 - UP

The idea in one sentence

We build an AI-powered digital librarian that reads studies on medicinal plants from Kosovo and the Albanian-speaking regions and, in seconds, produces a clear, sourced fact sheet for each plant.

Track 1 - Traditional knowledge

In simple words: you type a plant's name the way you say it, and you quickly get answers:

- which plant it really is
- how people use it and how they prepare it (for example as a tea)
- what scientists have found about it
- whether scientists tested the same kind of tea that people actually drink
- whether it carries a risk, or whether the plant itself is at risk of disappearing

Every line shows which study it comes from, so nothing is made up.

The problem

Knowledge about Albanian medicinal plants is scattered across many studies, and a student or researcher needs days to find a single answer.

In practice it works like this:

- Every study has different tables and formats.
- Albanian names change from area to area (dialects), so it is unclear which plant is meant.
- Each plant then has to be searched one by one in modern studies to see whether there is scientific evidence.
- People often don't know whether a plant carries a risk or is threatened by over-collection.

This matters because the knowledge is held by older people. The Sharr Mountains study relied on 139 elderly informants from Albanian, Bosniak/Gorani and Turkish communities and recorded 114 species used for medicinal purposes. Documenting this knowledge accurately, in a searchable form, is valuable before it is lost.

How it works

Five simple steps, all running automatically with the help of AI.

1. **We collect open studies** on medicinal plants from Kosovo and the Albanian-speaking regions.
2. **The system reads them** and pulls out the plant, the part used, how it is prepared and what it is used for.
3. **It links the Albanian name to the correct scientific plant**, even when the name is spelled differently or is a dialect form.
4. **It searches modern studies** for what was tested, and whether the same kind of preparation was tested (for example a hot tea, not an alcohol extract).
5. **It puts everything on one fact sheet**, with the risks, the conservation status and links to the sources.

The second trick: we give the system a new study it has never seen, and it adds the plant and its uses to the map by itself.

Where AI is used: to read unstructured text and turn it into a table. Where it is not used: for the important facts (risk, conservation status, level of evidence), which come from written rules and open databases so that they are accurate.

What we will use

In short: Python for the data work, an AI model to read the studies, public databases for the facts, and a simple web page for the demo.

Part	What we use	In one line
Data work	Python	Scripts that collect, clean and combine everything
Reading the studies	An AI language model (LLM)	Pulls plants, uses and preparations out of the text
Matching names	Spelling-tolerant matching and a plant-name database (GBIF or Plants of the World Online)	Links Albanian names to the right plant
Modern evidence	PubMed search	Finds the studies about each plant

Part	What we use	In one line
Risk and conservation	Open databases and written rules	Safety and at-risk flags come from rules, not AI guesses
Storage	A simple database	Keeps all the records
Demo	A simple web app with a map	What the judges see
Quality	GitHub, tests, a one-command rebuild	Anyone can reproduce our results
Checking	A hand-made sample and accuracy numbers	Proves it works

The data we will extract and use

In short: we turn facts buried in studies into a clean table, then add facts from open databases.

Data	Where it comes from	What we keep
Traditional use	Open studies from Kosovo and the Albanian-speaking regions	Plant, Albanian name, part used, preparation, use, source
Plant identity	Plant-name databases (GBIF, Plants of the World Online)	The correct scientific name for each plant
Modern evidence	PubMed studies	Which studies exist and what was tested (for example a tea or an alcohol extract)
Risk and safety	Published safety information	Known warnings and interactions, never doses
Conservation	Open databases such as IUCN	Whether the plant is at risk

Every record keeps a link to its source, and we write down each dataset's licence.

Other ideas and improvements

Ideas we could implement:

- [ ] A map of the Albanian-speaking regions showing where each use was recorded
- [ ] A time slider comparing old and new records
- [ ] A chat assistant that answers only from the fact sheets, with sources
- [ ] A conservation layer showing which plants are at risk
- [ ] Family recipes collected with consent
- [ ] A folklore and belief layer, kept separate from medicine
- [ ] A habitat map for the most at-risk plants